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IEEE PES TUNISIA Power & Energy Society

TSYP 14

IEEE TUNISIAN STUDENT AND YOUNG PROFESSIONAL CONGRESS

IEEE Industrial Electronics Society Tunisia Section Chapter

EnergyMind Tunisia (Architects of the Energy Noosphere: A Human-AI Collective Intelligence Network for Energy Resilience)

TSYP14 TECHNICAL CHALLENGE

SCOPE & TOPIC

The EnergyMind Tunisia challenge invites participants to build the trust layer of Tunisia's energy systems: an AI that detects when data or equipment is failing, faulty, or feeding the system false information, and decides transparently when it can act alone and when a human must decide.

Challenge problem

Tunisia's grid faces growing strain from:

- Seasonal peaks
- Fast-expanding renewable energy
- Increasing smart meters
- Growing numbers of IoT devices

These create more points where a fault, miscalibration or biased reading can silently corrupt the data used by automated systems.

As AI takes on more grid decisions, the question is not only whether it predicts well, but also:

Can it be trusted? Who is accountable when it is wrong?

The challenge is to design an AI system that stands between raw energy data and automated action.

Challenge Goal

Teams must design and build an AI-driven trust layer for a chosen part of the energy system:

- Microgrid
- Factory power line
- Smart-metered neighborhood
- Simulated grid segment

The solution must address four goals:

1. ANOMALY & BIAS DETECTION

Monitor incoming energy and grid data such as:

Voltage • Current • Power Quality • Smart-Meter Readings

Detect:

- Equipment faults
- Degraded sensors
- Biased readings
- Corrupted or falsified readings
- Natural equipment failures

The system must distinguish between:

Genuine Grid Event

e.g. real surge or outage & Data-Quality Problem

because they require different responses.

2. CONFIDENCE SCORES & EXPLAINABILITY

Every flagged event must provide a verdict that a human can understand and verify.

The system should:

- Output a confidence score
- Provide a plain-language explanation
- Maintain an auditable decision log
- Record what was detected, concluded and why

Teams are encouraged to use interpretable approaches such as:

SHAP • LIME • Rule-backed Models • Interpretable Ensembles

Avoid black-box scoring where the reasoning cannot be examined.

3. HUMAN-AI ESCALATION & GOVERNANCE

Clearly define what the AI can do independently and when a human operator must decide.

Teams should:

- Define thresholds for automatic corrective actions
- Define situations requiring human approval
- Provide an operator interface for rapid review, override or approval
- Address both over-trust and under-trust of AI

4. SHARED LEARNING ACROSS THE NOOSPHERE

Design and document how multiple nodes could share:

- Anomaly patterns
- Calibration insights
- Relevant knowledge

Nodes may represent:

Factories • Neighborhoods • Microgrids

The design must address privacy:

- What is shared?
- What remains local?
- Why?

A working single-node prototype is required. Multi-node sharing may remain a documented design.

Challenge Instructions

I. SCOPE DEFINITION

Each team defines the specific slice of the energy system it will monitor and designs the solution to address all four Challenge Goals.

II. DATASETS

Real fault/anomaly-labeled grid data is rare, so simulation is expected.

Teams may inject:

- Synthetic faults
- Sensor drift
- Biased readings
- Falsified readings

into real or realistic baseline data from:

- Public sources
- STEG open data
- Simulated equivalents

Teams must document how the anomalies were generated.

III. TECHNICAL INSIGHTS & HINTS

a. Anomaly Detection
- Isolation Forest
- Autoencoders
- One-Class SVM
- Statistical outlier methods

b. Explainability
- SHAP
- LIME
- Rule-backed scoring
- Readable decision logs

c. Escalation Logic
- Rule-based thresholds
- Confidence-based routing

d. Collective Defense
- Lightweight multi-agent concepts
- Federated concepts
- Documented sharing protocol

e. Evaluation
- Detection precision/recall
- False-positive rate
- Explanation quality
- Response latency

Phase 1: Ideation & Data Exploration

Required Deliverables

Required:

a. Project Proposal

Must include:
- Scope
- Objectives
- Data/anomaly-simulation plan
- Preliminary design for each Challenge Goal

b. GitHub Repository and/or Kaggle Notebook

Containing:
- Dataset
- Code developed during Phase 1

Phase 2: Solution Development & Deployment

Required Deliverables

a. Full Project Report

Covering:
- Complete workflow
- Escalation logic
- Collective-defense design

b. GitHub Repository

Containing:
- Full solution code
- Trained models
- Application files

c. Demo Video

Showcasing the working solution.: 5-Minute Presentation summarizing:
- Project
- Results
- Impact

Team Requirements

Maximum: 5 members per team

The challenge is open to Student Branches.

Pitching

English | 5 min presentation | 3 min demo

Scoring

TOTAL SCORE: 100 POINTS

Phase1: Pre-selection: 50 points

- Adherence to Challenge Requirements: 10 points
- Technical Approach: 20 points
- Quality & Clarity of Deliverables: 10 points
- Innovation and Creativity: 10 points

Phinal Phase: 45 points

- Technical Approach: 20 points
- Prototype Functionality & Features: 10 points
- Live Presentation Quality: 5 points
- Quality & Clarity of Deliverables: 10 points

Bonus: 5 points

- Having at least one PES member in the team: 1 point
- Having at least one IES member in the team: 1 point
- Having at least one YP member in the team: 1 point
- Having at least one member with both PES and IES membership; 1 point
- Covering the business scope — budget, solution implementation, production, continuous training, integration and deployment cost: 1 point

Important dates

- Challenge Infosession: **17/09/2026**
- Phase 1 Submission Deadline: **15/10/2026**
- Phase 2 Final Submission Deadline: **12/12/2026**

Prizes

3 WINNING TEAMS

Prizes: To Be Announced

The awards and prizes will be finalized and officially announced before TSYP14.

Submission

Submit your project and show us what you've built.

Contact



Please contact us through the following email address ranimdamak@ieee.org or mohamedali.krichen@ieee.org for any inquires or clarifications.